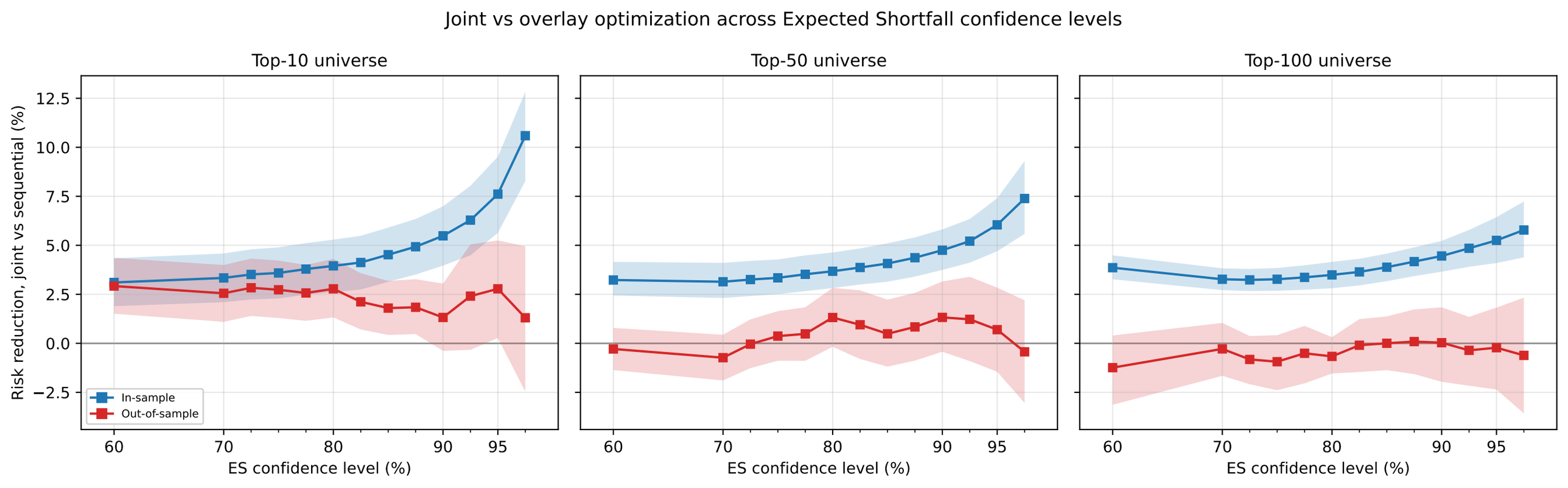
Joint vs overlay optimization across Expected Shortfall confidence levels
Top-10 universe
Top-50 universe
Top-100 universe
Risk reduction, joint vs sequential (%)
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
In-sample
Out-of-sample
60
70
80
90
95
ES confidence level (%)